ROOTACCESS PLATFORM TEST PLAN & TEST SPECIFICATION

Project Target:	RootAccess Live Platform (https://rootaccess.co.in)
Test Framework:	Playwright Test (JavaScript, Page Object Model, ES Modules)
Document Type:	Master Test Plan (MTP) + Standard Test Case & Defect Formats
Version & Date:	Version 1.0 | October 2026 | Status: APPROVED & ACTIVE

1. EXECUTIVE SUMMARY & TEST OBJECTIVES

The primary objective of this Test Plan is to define the testing strategy, architecture, scope, test case formats, defect lifecycle standards, and verification procedures for the RootAccess practical engineering learning platform.

- **Defect Prevention:** Identify regressions, UI defects, and broken user journeys early in the development lifecycle using automated smoke and regression test suites.
- **Industry-Standard Design:** Implement a clean, beginner-friendly but production-grade Page Object Model (POM) using modern Playwright APIs (web-first assertions, auto-waiting, accessible locators).
- **Cross-Browser & CI/CD Readiness:** Enable seamless test execution across local workstations and headless CI/CD pipelines (GitHub Actions), generating HTML reports, traces, and screenshots.

2. TEST SCOPE: IN-SCOPE VS OUT-OF-SCOPE

2.1 In-Scope Modules & Features:

- **Homepage (Public):** Brand hero, navigation links, course track catalog cards, interactive FAQ accordions, marketing callouts, and footer links.
- **Authentication & Security:** Sign-in form validation (empty fields, invalid credentials), password visibility toggles, and protected route access control guards.
- **Course Curriculum Explorer:** Course syllabus overview, milestone trackers, learning outcomes, and syllabus PDF download actions.
- **Lesson Viewer & Interactions:** Lesson content rendering, code snippets copy actions, practical QA execution checklist items, and previous/next navigation.
- **Certification Quizzes:** Assessment loading, question & option rendering, single-best answer selection, score calculation (75% passing threshold), and technical explanation reviews.
- **Learner Progress & Persistence:** Overall completion percentages, milestone counters, and local storage state persistence.

2.2 Out-of-Scope:

- **Destructive Operations:** Dropping backend database tables or modifying production student records.
- **Third-Party Email Handshake:** Direct verification of external email delivery services (handled via mock tokens/stubs in staging).

3. TEST STRATEGY & FRAMEWORK ARCHITECTURE

RootAccess employs a lightweight, professional Page Object Model (POM) architecture using pure JavaScript (ES Modules). Tests are strictly decoupled from DOM locators to maximize maintainability.

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PROJECT TEST REPOSITORY STRUCTURE:
|-- pages/                # Reusable Page Objects (HomePage.js, LoginPage.js, CoursesPage.js)
|-- tests/                # Feature Test Suites (home.spec.js, login.spec.js, courses.spec.js, ...)
|-- test-data/            # Reusable non-sensitive fixtures & data (testData.js)
|-- playwright.config.js  # Central Playwright runner & browser configuration
|-- playwright-report/    # Self-contained interactive HTML test reports
```

3.1 Playwright Locator Precedence (User-Centric & Accessible):

- **Priority 1:** getByRole() - Tests interactive accessibility semantics (e.g. button, heading, link, alert)
- **Priority 2:** getByLabel() - Binds to form controls through visible labels
- **Priority 3:** getByPlaceholder() - Targets input fields with explicit placeholders
- **Priority 4:** getByText() - Asserts visible copy, headings, and descriptions
- **Anti-Pattern:** Avoid fragile XPaths, deep CSS chains (#app > div > div:nth-child(3)), or arbitrary timeouts.

3.2 Auto-Waiting & Web-First Assertions:

All tests leverage web-first assertions (e.g. `await expect(locator).toBeVisible()`). Playwright automatically polls the DOM until elements become visible, attached, and stable, completely eliminating hardcoded `page.waitForTimeout()` flakiness.

4. TEST ENVIRONMENT & RUN CONFIGURATION

Parameter	Local Development	CI / GitHub Actions	Notes
Target Base URL	http://localhost:5173	https://www.rootaccess.co.in	Configured via BASE_URL
Browser Engine	Chromium (Desktop)	Headless Chromium	Expandable to WebKit/Firefox
Parallel Workers	4 Local Workers	1-2 CI Workers	Safe parallel execution
Default Timeout	30,000 ms	30,000 ms	Global test execution timeout
Expect Timeout	5,000 ms	5,000 ms	Assertion auto-wait timeout
Failure Artifacts	Trace + Screenshot	Full Trace + Screenshot + Video	Retained on failure

5. INDUSTRY-STANDARD TEST CASE FORMAT & SPECIFICATION

Below is the standard test case specification format adopted for RootAccess quality assurance, aligned with IEEE 829 and enterprise ALM tools (e.g., Jira, TestRail, Zephyr).

TEST CASE SPECIFICATION: TC-HOME-001 (HOMEPAGE HERO & NAVIGATION)

Test Case ID	TC-HOME-001
Module / Feature	Homepage | Marketing & Navigation
Test Priority	P1 - High | Type: Functional / Smoke
Pre-conditions	Browser is launched; base URL is accessible and responsive.
Test Data	N/A (Public marketing landing page)
Execution Steps	1. Navigate to "/" -> Verify RootAccess logo is visible 2. Verify H1 headline renders with brand text
Expected Result	Homepage renders in under 2s; all navbar links are clickable; hero headline matches brand message.
Actual Result	Verified Passed via tests/home.spec.js (Automated Playwright test execution).
Automated Status	PASS (Execution duration: 7.6s, automated)

TEST CASE SPECIFICATION: TC-AUTH-001 (REQUIRED FIELD VALIDATION)

Test Case ID	TC-AUTH-001
Module / Feature	Authentication | Sign In Page
Test Priority	P1 - High | Type: Negative / Boundary
Pre-conditions	User is logged out; session storage has no active auth token.
Test Data	Empty email (""), empty password ("")
Execution Steps	1. Navigate to "/login" 2. Leave email and password empty
Expected Result	Alert displays "Please enter your email address." with role="alert"; no network call sent.
Actual Result	Form prevents submission and displays accessible error banner.
Automated Status	Automated in login.spec.js

TEST CASE SPECIFICATION: TC-QUIZ-001 (ASSESSMENT SCORING & THRESHOLD)

Test Case ID	TC-QUIZ-001
Module / Feature	Assessment Engine | Certification Quiz
Test Priority	P2 - Medium | Type: Functional / Business Logic
Pre-conditions	User is enrolled; navigated to "/courses/manual-qa-fundamentals/quiz".
Test Data	Questions 1–4 correct answers: [1, 2, 1, 0]
Execution Steps	1. Select option B for Q1, option C for Q2, option B for Q3, option A for Q4 2. Click "Submit & View Results"
Expected Result	Score calculated as 100%; >= 75% benchmark triggers passing banner and unlocks certificate.
Actual Result	Score calculated deterministically; results saved in progress context.
Automated Status	Automated in quiz.spec.js

6. INDUSTRY-STANDARD DEFECT / BUG REPORT FORMAT (JIRA STANDARD)

When an automated or exploratory test detects unexpected application behavior, defect reports must follow this standardized template to ensure rapid triage, reproducibility, and resolution.

DEFECT REPORT TEMPLATE (JIRA / BUGZILLA COMPATIBLE)

Defect ID	BUG-ROOT-[ID] (e.g. BUG-ROOT-042)
Defect Title	[Module] Short summary of behavior under specific condition
Severity vs Priority	Severity: Critical | Major | Moderate | Minor Priority: P1 (Blocker) | P2 (High) | P3 (Medium) | P4 (Low)
Environment	OS: macOS Sonoma 14.x | Browser: Chromium 133.0 | URL: http://localhost:5173
Pre-conditions	Exact account state, local storage keys, or prerequisites needed
Steps to Reproduce	1. Step one (Action) 2. Step two (Input/Navigation)
Expected Result	What the application should have done according to specifications
Actual Result	What actually occurred (including error codes, blank screens, or UI bugs)
Evidence / Artifacts	Playwright Trace (.zip), Failure Screenshot (.png), Video (.webm), Console Logs
Root Cause Analysis	Identified component / hook / route causing the defect (filled by dev/QA)

7. ENTRY, EXIT & SUSPENSION CRITERIA

7.1 Test Entry Criteria:

- **Code Freeze & Build:** Vite build compiles cleanly (`npm run build` exits 0 with zero TypeScript errors).
- **Environment Health:** Target URL responds with HTTP 200; database connectivity verified via `npm run db:verify`.

7.2 Test Exit Criteria:

- **100% Smoke Pass Rate:** All core P1 smoke tests (home, login, course exploration) pass without failure.
- **Zero Blocker Defects:** No open Critical/P1 defects blocking main student user journeys.
- **Report Publication:** HTML test execution report archived with timestamped test execution summary.

7.3 Test Suspension & Resumption Criteria:

- **Suspension:** Triggered if local dev server fails to launch or database authentication service is unreachable.
- **Resumption:** Testing resumes once service is restored and database health check passes.

8. QA ENGINEER COMMAND REFERENCE

- **Run Full Suite:** `npm test` (Executes all Playwright tests headlessly)
- **Run Headed Mode:** `npm run test:headed` (Executes tests with visible Chromium browser)
- **Interactive UI Mode:** `npm run test:ui` (Opens time-travel debugging & locator picker)
- **View HTML Report:** `npm run test:report` (Opens self-contained HTML test report in browser)
- **Inspect Unit Tests:** `npm run test:unit` (Runs isolated Vitest unit test suite)